

Value-Aware Team-Level Stopping for Multi-Robot Exploration using Hierarchical MARL

MARL Research Proposal

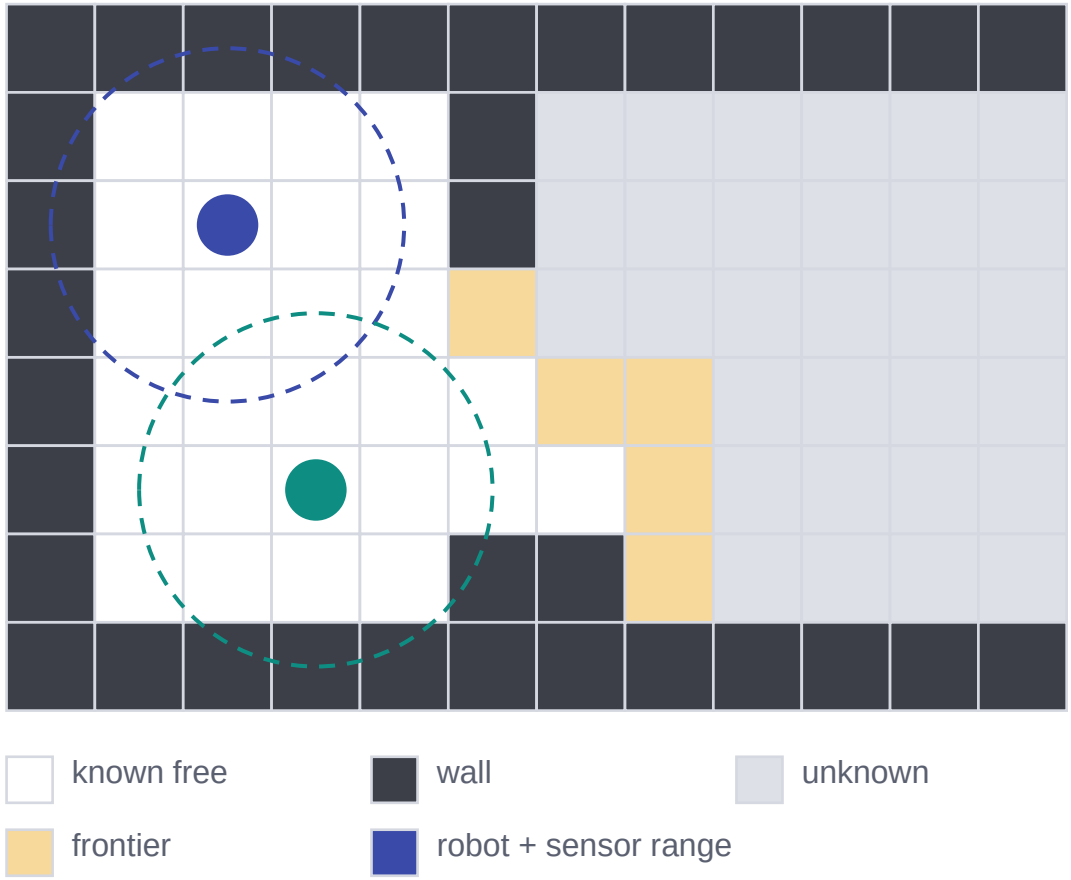
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Multi-robot exploration



Occupancy grid. The map is a grid of cells: free, wall, or unknown. Robots start knowing nothing.

Sensing. Each robot sees cells within a fixed radius, in line of sight. Walls block the view.

Frontier. A known free cell next to an unknown one. Going there reveals new space [1].

Allocation. Each robot gets a different frontier, trading off distance and expected gain [2].

Exploration cycle. Sense → share maps → assign frontiers → move. Repeat.

The open question. After how many cycles should the whole team stop?

How exploration stops today

Stopping rule	How it decides	Used in	What it ignores
No frontiers left	Explore until nothing reachable is unknown	Yamauchi 1997 [1]; Burgard et al. 2005 [2]	Cost of the last, scattered pockets
Coverage or time budget	Stop at X% mapped or after T minutes	Common in practice and in MARL benchmarks	X and T must be re-tuned for every map and team
Uncertainty saturation	Stop when map and pose uncertainty stop improving	Placed & Castellanos 2022 [3]	Single robot; motion and messages not priced
Learned map completeness	A CNN judges if the partial map is complete	Luperto et al. 2026 [4]	Single robot; supervised, not cost-aware
Per-robot return rule	A robot relays data when returning pays more than exploring	Kim et al. 2026 (PRoID) [5]	Per robot and hand-set; the mission ends at a deadline

Common thread: no rule decides, for the whole team, whether one more cycle is worth what it costs.

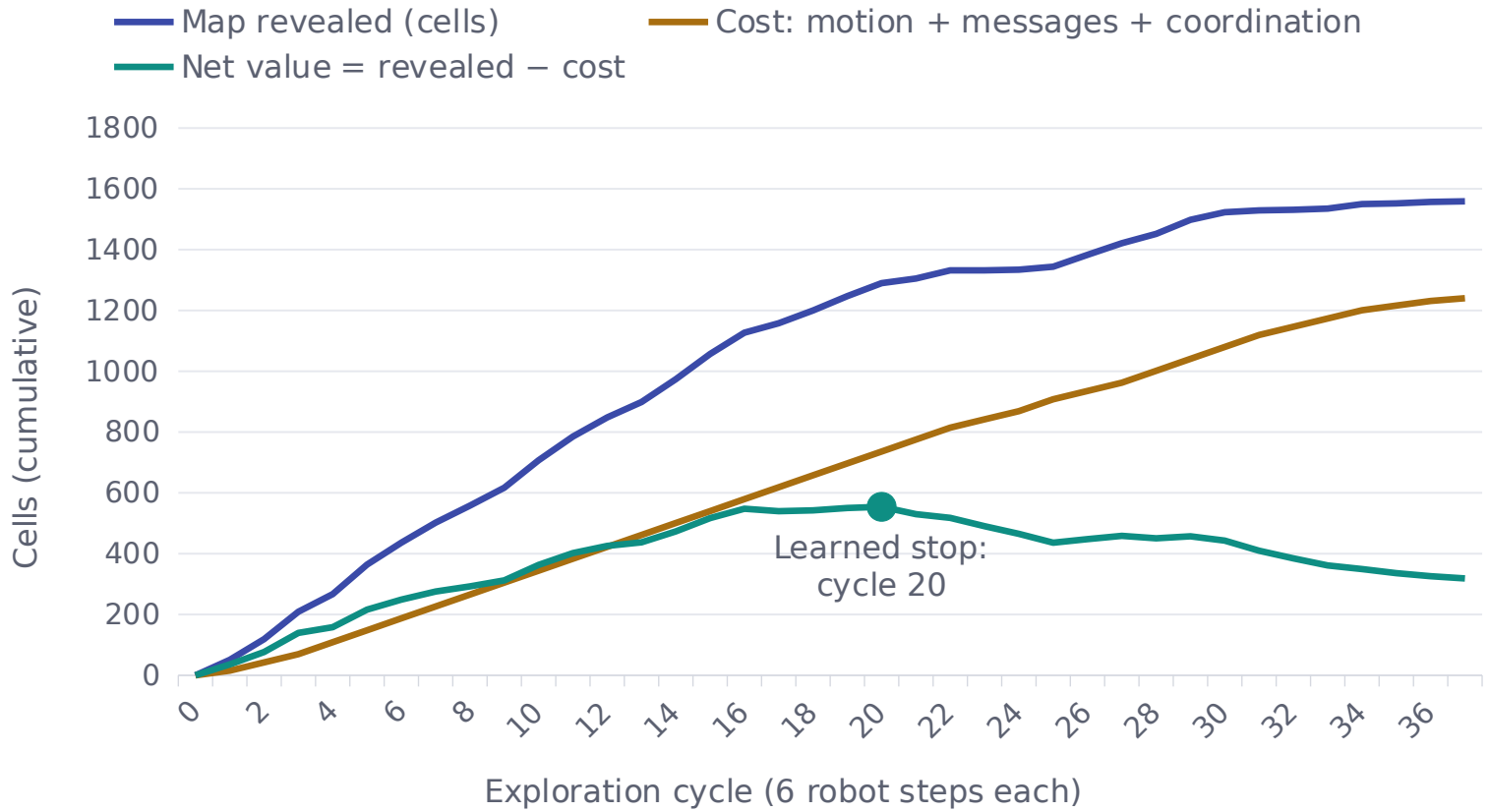
When should the team stop?

Early cycles reveal whole rooms. Late cycles reveal small, scattered pockets that need long drives.

Every cycle keeps costing: robot motion (energy, time), map-sharing messages, re-planning.

So the question is not whether the team *can* explore further, but whether one more cycle is **worth its cost.**

Messages grow as N(N−1) with team size N, so the question gets sharper for bigger teams.



Prototype run, 3 robots, 40×40 map. Stopping at cycle 20 maps 83% of the free space; the last 17 cycles add 17% more map but cut net value from 554 to 319.

Three problems, one contribution

Problem A

Frontier allocation

Which robot explores which frontier?

Well studied: greedy [2], market-based [6], MinPos [7], and recent MARL methods (Tan et al. 2023 [8]; MARVEL 2025 [9]).

Execution substrate

Problem B

Communication

Which robots talk, and what do they share?

Active MARL topic: learned, targeted and bandwidth-limited messaging (TarMAC 2019 [10]).

Modelled as a cost and a constraint

Problem C

Team-level stopping

When is one more cycle no longer worth its cost?

Exploration uses fixed or single-robot rules [1]–[5]; MARL work fixes the episode length [8], [9].

The contribution of this project

Frontier allocation is the machinery the robots run on, not the claim. **The claim is about when the whole team should stop.**

What is already known

Area	What exists	What is still missing
Multi-robot exploration Yamauchi 1997 [1]; Burgard et al. 2005 [2]; MinPos 2012 [7]; survey [11]	Frontier-based coordination with utility and travel-cost allocation.	Termination is a fixed rule: no frontiers left, coverage or time.
MARL for exploration Tan et al. 2023 [8]; MARVEL 2025 [9]	Learned, decentralised frontier selection; CTDE training.	Episodes end at a fixed length or coverage; stopping is not learned.
Stopping criteria for exploration Placed & Castellanos 2022 [3]; Luperto et al. 2026 [4]; PRoID 2026 [5]	Uncertainty-based and learned single-robot stop rules; per-robot return rules.	A team-level, learned, cost-aware decision to end the mission.
Hierarchical MARL Options, Sutton et al. 1999 [12]; Han et al. 2019 [13]	Learned termination of each agent's sub-policy (option).	Termination of the whole task, driven by expected gain versus team cost.
Optimal stopping Tsitsiklis & Van Roy 1999 [14]; Longstaff & Schwartz 2001 [15]; Becker et al. 2019 [16]	Theory and regression or neural stopping rules, mostly single decision maker.	Many agents, partial observability, costs that depend on coordination.
LLM-agent orchestration Hu et al. 2025 [17]; Zhang 2026 [18] (motivation only)	Statistical stop rules for LLM debates; a 2026 preprint survey finds no RL-trained stop.	Same question in a different field. Not part of the proposed system.

Closest prior work, and the gap

Work	Multi-agent	Spatial exploration	Learned stop	Ends the whole mission	Prices motion + messages + coordination
Placed & Castellanos 2022 [3]	✗	✓	✗	single robot	✗
Luperto et al. 2026 [4]	✗	✓	supervised	single robot	✗
Kim et al. 2026 (PRoID) [5]	✓	✓	✗	per robot	partly
Szostak & Cohen 2024 [19]	✓	✗	✓	partly	sampling cost
Han et al. 2019 [13]	✓	✗	✓	sub-policies only	✗
This project	✓	✓	✓	✓	✓

Szostak & Cohen [19]: multi-agent RL that learns when to stop sampling in hypothesis testing (no map, no motion). Han et al. [13]: learned termination of options, not of the task.

In the prototype the stop is learned by one central learner; the MARL version is the semester's work.

Research questions

Can a hierarchical MARL policy learn a team-level stopping strategy that balances expected information gain against motion, communication and coordination costs during multi-robot exploration?

RQ1 **Can MARL learn efficient frontier allocation?**

Level-2 policy vs greedy and market-based allocation, with stopping held fixed.

RQ2 **Can the policy tell when further exploration has low marginal value?**

Learned value vs the value known in hindsight; stop time vs the best stop in hindsight.

RQ3 **Does learned stopping save time and travel without losing meaningful coverage?**

Net value, coverage at stop, distance and messages against every baseline.

RQ4 **How does it scale with team size and communication limits?**

Train on 2 robots; test on 2 to 10 robots, 3 map sizes, 3 communication levels.

Formulation 1: the setting

Symbol	Meaning	In this project
I = {1, …, N}	The agents	N robots (trained with 2, tested with 2 to 8, later 10)
S	True state of the world	True map, robot positions, what has been mapped so far
Ω_i, O	What robot i observes	Cells within 4 cells of the robot, in line of sight
A_i	Robot i's action (Level 2)	Which frontier to drive to, one step at a time
A_0	Team action (Level 1)	Continue or stop, once per cycle; later also: how many robots stay active
P	How the state changes	Robots move and sense; the known map grows
R	Reward	Net value of each cycle: cells revealed minus the team's costs (next slide)
b	Belief / team summary	What the team knows at a decision point, compressed into features

Dec-POMDP [20]: M = ⟨ I, S, {A_i}, P, R, {Ω_i}, O ⟩

Team decision (Level 1): $a_0(t) \in$ { continue, stop }, taken once per cycle from the team summary b(t)

Dynamics and observations: P(s′ | s, a_1…a_N) and O(o_1…o_N | s′, a_1…a_N)

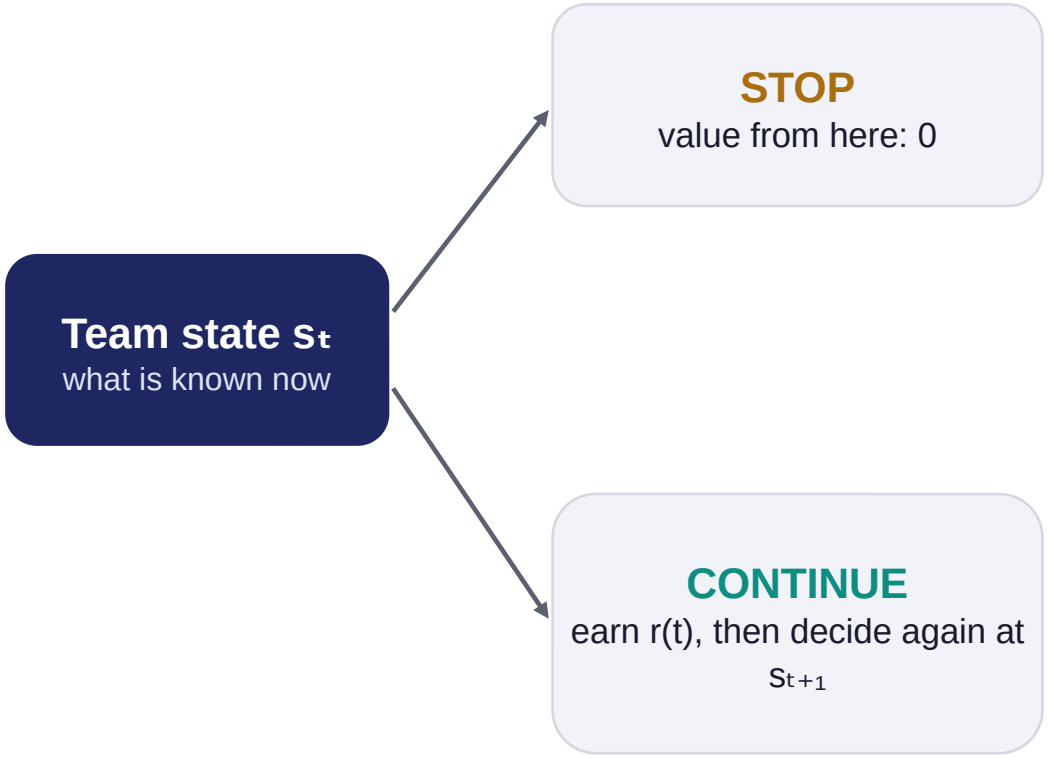
Formulation 2: value of a cycle

ΔI(t)	cells revealed in cycle t (new map)
d(t)	robot steps actually taken in cycle t
N(N−1)	map-sharing messages: every robot to every other
c_coord	fixed re-planning overhead per cycle

Stopping after T cycles earns V(T) = r(0) + … + r(T−1).

$$r(t) = \Delta I(t) - c_{move} \cdot d(t) - c_{comm} \cdot N(N-1) - c_{coord}$$

Formulation 3: optimal stopping



Value of continuing

$$Q(s) = E[\, r(t) + \max(0, Q(s_{t+1})) \mid s_t = s \,]$$

max(0, ·): at the next decision the team again picks the better of stop (0) and continue. This is the classical optimal-stopping recursion [14], [15].

Stopping rule

Stop at the first t with $Q(s_t) \leq 0$

i.e. when the expected value of exploring further no longer covers its cost.

Reference for scoring (not a policy)

$T^* = \text{argmax}_T V(T)$, computed after the run

Needs the future. Every rule is scored as a % of $V(T^*)$.

How the stop is learned

1 Collect

300 training maps, 2 robots, 32×32. Explore each to the end and log every cycle.

2 Re-score

Score every run under 4 random cost settings (cost randomisation): 1,200 runs.

3 Fit Q

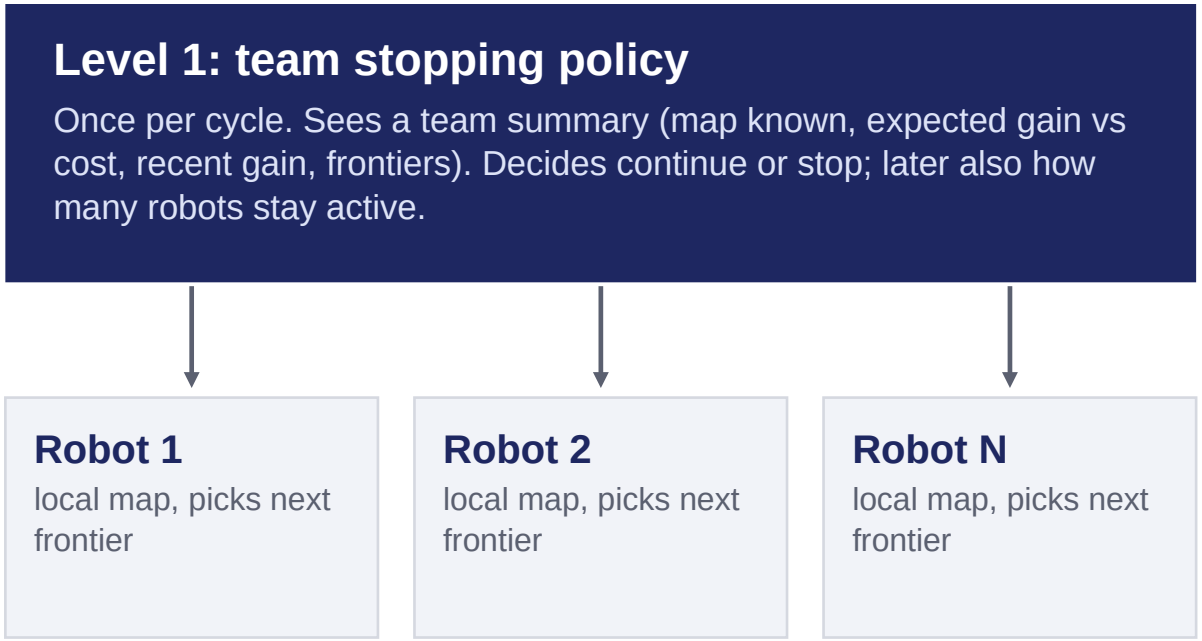
25 rounds of fitted Q-iteration [21]; each round an extra-trees regressor (80 trees).

4 Decide

At run time compute Q from the 7 features. Stop when Q ≤ 0.

Feature (all ratios, so they transfer)	Meaning
Map known	Fraction of all grid cells already known
Expected gain ÷ cycle cost	Unknown cells near the planned targets, per unit of cost
Last gain ÷ cost, previous gain ÷ cost	How well the last two cycles paid off
Distance to targets ÷ 6	How far away the planned frontiers are
Share of robots with a target	How much of the team still has work
Frontier cells ÷ map side	How much frontier is left

Two-level architecture



Level 2: one policy per robot, every step, shared parameters (planned: MAPPO [22]; QMIX [23] as backup)

CTDE [20], [22], [23]: trained with the full map and all robot states; executed with each robot's own observations. That is what makes it multi-agent hierarchical RL [12], [13].

Baselines for comparison

Stopping rule	Stops when…	Parameter	Idea
Explore until no frontiers	no reachable frontier is left	none	Classical exploration [1], [2]
Fixed budget	B cycles have run	B = 19 (tuned)	Time budget
Coverage threshold	known fraction of the grid ≥ τ	τ = 0.84 (tuned)	Area threshold
Myopic rule	the last cycle lost value (r < 0)	none	Stop at the first bad cycle
Gain-vs-cost rule	α × expected gain ÷ cycle cost < 1	α = 1.6 (tuned)	Hand-made value test
Learned value-aware stop	estimated Q ≤ 0	learned	Looks at the whole remaining future
Best in hindsight (oracle)	best cycle, chosen after the run	needs the future	100% reference, not a policy

With α = 1.6 the gain-vs-cost rule stops once expected gain < cycle cost ÷ 1.6 (about 0.63 × cost).

Evaluation metrics

Net value, % of best

V(T) ÷ V(T*), totals over all maps

Main score: did it stop at a good time?

Regret

V(T*) − V(T), as % of the map

Value lost; stable even when V(T) is tiny*

Stop-time error

T − T* in cycles; share early / late

How and when it gets it wrong

Operational cost

Coverage, cycles, robot steps, messages

What the stop saves, and what it leaves

Value estimate quality

Correlation of Q with hindsight value; sign accuracy

Does it know the remaining value?

Robustness

Worst setting, interquartile mean, settings ≥ 80%

Does it hold without retraining?

Statistics: paired comparison on identical maps, 95% bootstrap intervals, and (for MARL) at least 5 training seeds, following Agarwal et al. [24].

Phased plan

Phase 1

Toy environment

N agents, M tasks, communication cost, a stop action. Does the policy learn when to continue, stop, and how many agents to activate?

Phase 2

2D grid-world exploration

Partial maps, frontiers, sensing, team-size and communication tests. Most quantitative results come from here.

Phase 3

ROS2 / Gazebo

MRTA testbed: realistic motion and sensing. Checks that the policy transfers to a real robotics stack.

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Thank you

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